Weak Choice Principles for Proof-Relevant Setoids

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CCS Concepts: • **Theory of computation** → **Type theory**.

Additional Key Words and Phrases: type theory, dependent types, rewriting theory, confluence, termination

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1 INTRODUCTION

Writing formal proofs in dependent type theory is a delicate art, even for the most seasoned users of interactive proof assistants: the challenge often lies not so much in working out the proof itself, but in finding the right definition to build upon. This is especially true in *intensional* type theory [Martin-Löf 1971], where the equality judgment only identifies those definitions that are $\beta\eta$-convertible. Even though the intensional character of the judgmental equality can seem overly restrictive compared to the extensional nature of the mathematical equality, it is in fact a necessary condition for the decidability of the typing relation, which is important for both foundational and practical reasons.

The intensional nature of the equality judgment serves an important purpose, but the same cannot be said of the *propositional* equality, which happens to be just as coarse. As a result, users commonly postulate extensionality axioms for the propositional equality, such as the principle of function extensionality, which states that two functions are equal if their images are equal, the principle of proposition extensionality, which states that logically equivalent propositions are equal, the principle of uniqueness of identity proofs (UIP), or even the univalence axiom. Unfortunately, axioms tend to break the canonicity of the system: function extensionality will result in closed terms of type $\mathbb{N}$ that do not evaluate to an actual integer.

An alternative strategy is to work with types that are equipped with an equivalence relation, which are known as *setoids*. Doing so allows us to specify the meaning of equality for every type, but it comes at the cost of having to manually prove that every function preserves the setoid equality. This should be the work of a compiler, not of the user! What we really want is a way to interpret type theory with extensionality principles into plain intensional type theory. The first such interpretation was given by Hofmann [1995c] with his setoid model, which interprets all types as setoids and validates the principles of function extensionality, proposition extensionality and UIP. Unfortunately, Hofmann's setoid model only interprets a fragment of type theory, in particular

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it does not support universes or large elimination (the ability to define types by pattern-matching on values) which makes it impossible to even prove that $0 \neq 1$.

[Its spiritual successor is observational type theory, which has been justified both by translations or by logical relations. But OTT is definitionally proof irrelevant, (here I need to talk about having a specific sort for propositions) which means that it does not support Acc-elim. Plus, it does not have unique choice.]

Contributions. In this paper, we construct a new universe of setoids which does not require the equalities to be definitionally proof-irrelevant. We then use our universe of setoids to construct two extensions of Hofmann's setoid model: a *predicative* model which uses Type-valued equalities and can be formulated in plain Martin-Löf Type Theory (MLTT), and an *impredicative* model which uses Prop-valued equalities and can be formulated in the Calculus of Inductive Constructions (CIC). The predicative model supports all the type formers of MLTT plus quotient types, and validates function and proposition extensionality, as well as the principle of unique choice. The impredicative model adds support for impredicative quantification and large elimination of the accessibility predicate, but it does not validate the principle of unique choice.

Because our models are constructed in plain intensional type theory, they show that extensionality principles and weak choice principles do not add any consistency strength to MLTT or CIC. By combining this result with the conservativity of extensional type theory over intensional type theory with extensionality principles [Hofmann 1995a; Winterhalter et al. 2019], we obtain that the same is true for the equality reflection rule. More generally, any arithmetical theorem that can be proven in MLTT or CIC extended with extensionality principles, weak choice principles or the equality reflection rule can also be derived without them.

Plan of the paper. The usual.

1.1 Related work

The setoid model was originally conceived by Hofmann [1995b] in order to extend intensional type theory with extensionality principles and quotient types. However, this first iteration did not support universes or large elimination. Building on this work, Altenkirch [1999] used a type theory extended with a sort of definitionally proof-irrelevant propositions to give a stricter version of the setoid model, and equipped it with a universe of simple types which contains only non-dependent function types. This already provides a limited form of large elimination, and in particular this is enough to prove that $0 \neq 1$. Finally, Altenkirch et al. [2021] study several methods to construct a universe that supports dependent type formers, using further extensions of the ambient type theory such as induction-recursion, induction-induction or a strengthened J eliminator for the equality. Compared to their work, our construction only needs plain indexed inductive types – in fact, it does not even require definitional proof-irrelevance.

Meanwhile, Altenkirch et al. [2007] explored a different way to mix extensionality and intensionality with Observational Type Theory (OTT), an intensional type theory whose equality type is replaced with a definitionally proof-irrelevant setoid equality. This strategy allows them to implement quotient types and extensionality principles, while preserving the good computational properties of their system. However, Sterling et al. [2022] observe that the utility of quotient types is substantially diminished in the absence of unique choice. Subsequently, the community explored several points in the design space of observational type theories: XTT [Sterling et al. 2022] uses a cubical syntax to implement extensionality principles, and implements universe induction, but still lacks unique choice. $\mathsf{CIC}^{\mathsf{obs}}$ [Pujet and Tabareau 2022] adds an impredicative sort of propositions and a scheme of indexed inductive definitions to bring their system closer to CIC, but they are not able to implement large elimination of the propositional accessibility predicate. Our work is able

to handle unique choice for the predicative model, and large elimination of accessibility for the impredicative model.

2 HOFMANN'S SETOID MODEL

2.1 Our metatheory

- The usual CIC with indexed inductive families
- We will need an impredicative Prop only if we want impredicativity

2.2 Hofmann's model

- Blah blah

3 DEFINITION OF THE UNIVERSE

3.1 Inductive-Recursive intuition

- The fundamental idea is that we want to define the universe inductively, while defining the setoid equality by recursion
- That is technically double small IR
- in Kaposi et al the suggest to eliminate via induction-induction

3.2 A Plan in Three Steps

- Start like you would with small IR
- Except that here we are trying to do double IR, which is technically not small
- Therefore, we use this weird encoding that still somehow works
- Talk about the elimination principle that we get for the universe
- Probably it has the right computation rules? Maybe look at strong vs weak in Kaposi et al

3.3 Transitivity and Coercion

- Mutually define transitivity, coercion, and the equality of coercion
- Do it forward and backward simultaneously so we can pass Coq's guard condition

3.4 Second universe and embedding

- Now that the universe is a proper setoid, we can do a second universe level
- Put an embedding constructor, rinse and repeat
- Obviously we can get a full hierarchy
- No cumulativity though :shrug: maybe we can define emb as a function instead, which might be a tad bit cooler?

4 W TYPES AND RECURSION-RECURSION

4.1 W Types and their equality

- We have to restrict a tiny bit the sup to only use extensional functions
- The equality is inductively generated

4.2 Eliminating W Types

- We need to define the eliminator while simultaneously proving it extensional
- Not so easy actually! The natural way to do it is via double recursion-recursion
- While simple recursion-recursion is trivial to eliminate (just do a sigma lol), double recursion recursion is more technical

4.3 Graph encoding

- Define the graph inductively
- Show that it is extensional
- Then show that you can always find

5 QUOTIENTS AND UNIQUE CHOICE

5.1 Adding Quotient Types

- Not very difficult tbh, just add them

5.2 The Principle of Unique Choice

- Allows for effective quotients
- Logically very powerful, very close to replacement
- Impossible for proof-irrelevant OTT, since there's no info in equality
- Unfortunately the naive version causes some universe issues
- Maybe we can use Pédrot's trick
- But for that we would need a proper normalisation proof for an impredicative theory

5.3 A Predicative Theory

- Instead, we propose a new model with a predicative hierarchy of propositions
- That way, we can get a principle of unique choice which actually works
- In this model, functions and functional graphs are exactly the same

6 IMPREDICATIVE QUANTIFICATION AND ACCESSIBILITY

6.1 Flirting with Russell's paradox

- recall why impredicativity is nice?

6.2 Well-founded induction and Weak Choice Principles

- Explain how acc elimination works
- Allowed in Lean with strict proof irr, failure of normalisation
- Disallowed in OTT, which implies that we cannot extract a function that normalises System F
- It implies some small bits of choice!

6.3 Adding Acc elimination

- We need to do the graph trick again
- but it works, nice!

7 CONCLUSION AND FUTURE WORK

- Recall the results: we only need inductive family for our model of extensionality!
- Maybe some applications stolen from Hofmann's thesis?
- Maybe make it into a proper type theory, OTT style
- We are not going to have cast on refl, though :shrug:
- Normalisation proofs, normalisation proofs, normalisation proofs
- Making unique choice work with impredicativity?

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